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equivR: Reproducible Cross-Runtime Equivalence for R Analytics

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Executive Summary

R is becoming an important open-source option for clinical-trial analysis and regulatory submissions, but long-term reviewability remains a practical challenge as packages, runtimes, and dependencies change. Environment preservation helps, but some analytical workflows eventually need to be refactored or migrated rather than frozen in place.

equivR addresses that next step: modernize trusted analytical code while independently verifying that its analytical behavior is preserved.

Generation proposes; validation decides.

Phase 1 will first demonstrate SAS -> R modernization using pregenerated trusted SAS reference artifacts, then apply the same workflow to legacy R -> modern R. Over four months, the project will deliver an open-source **equivR** core under MPL-2.0 with deterministic equivalence validation, structured diagnostics, a model-independent LLM repair loop, R-friendly local/CI workflows, and reproducibility evidence.

Signatories

Project team

Yonglin Zhu — co-PI / co-lead. Led a completed SAS-to-Python modernization framework combining AI-assisted transformation with numerical-equivalence validation. Leads architecture, contract design, and the validator.

Xing Cai — co-PI / co-lead. Founding member of the R Working Group at PPD, working across SAS and R analytics. Leads use-case design, benchmarks, testing, and documentation.

equivR is independent open-source work, not sponsored, funded, or endorsed by either co-PI's employer, and does not use employer-owned code, data, or infrastructure.

Contributors

The co-PIs have already prepared this proposal and a public prototype at <https://github.com/zhuylu/equivR>: a base-R proof of concept with worked examples, a Phase 1 implementation plan, and early contract-v1 validation code with tests and CI. Community contributions will be invited through the repository.

Consulted

The design is informed by public R Consortium Submissions Working Group materials (R Consortium 2026a, 2026b, 2026c); citation does not imply endorsement.

The Problem

Recent R Consortium discussion on making R submissions reviewable for FDA highlights a practical challenge: reviewers may receive R code but still struggle to recreate the sponsor’s environment, install dependencies, or reproduce submitted results (R Consortium 2026a). As R becomes a more important open-source option for clinical-trial analysis, reviewability and reproducibility matter as much as whether the code runs.

The R Submissions Working Group is addressing this through approaches such as containers and WebAssembly, while also planning greater use of AI and automation in future submission pipelines (R Consortium 2026b).

Preserving an old environment does not solve every long-term problem. R versions and packages change, legacy code becomes hard to maintain, and organizations eventually need to refactor or migrate analytical workflows. The question then becomes:

Can we change the implementation while preserving the analytical behavior that made the original workflow trustworthy?

Today, teams usually answer this with project-specific comparison scripts and manual review. There is no common infrastructure for defining what must remain equivalent and checking it across legacy R, modern R, SAS, Python, or AI-assisted rewrites.

Large language models make modernization easier, but they cannot be trusted to validate their own output. **equivR** targets this missing layer: LLM-assisted modernization combined with independent equivalence validation, so analytical workflows can evolve while remaining reproducible, reviewable, and trustworthy.

The proposal

Overview

equivR will let teams modernize analytical code while keeping the trusted implementation as the authority for what must not change.

The first target is SAS -> R. A trusted SAS workflow provides reference artifacts; an LLM produces or repairs an R implementation; **equivR** validates the R outputs against the SAS reference and returns structured diagnostics on failure. A candidate is accepted only when all declared checks pass. The same framework is then applied to legacy R -> modern R.

For the R community, this offers a second path to long-term reviewability: instead of preserving an old runtime indefinitely, a workflow can be modernized with verifiable evidence that its analytical behavior is unchanged.

Detail

The project separates code generation from acceptance: an LLM may propose a refactor, translation, or repair, but only deterministic `equivR` validation decides whether the result preserves the trusted analytical behavior (Figure 1).

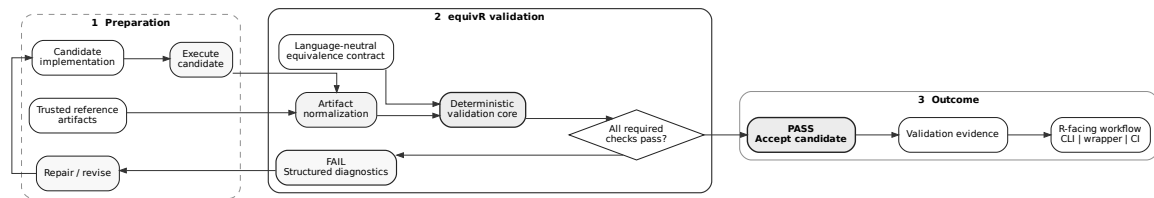


Figure 1: LLM-generated candidates are accepted only after deterministic validation against trusted reference artifacts.

Minimum Viable Product

The Phase 1 MVP will let a user:

1. provide trusted reference artifacts and an R candidate;
2. define an equivalence contract (keys, schema, missingness, categorical values, numerical tolerances, selected analytical results);
3. run deterministic validation locally or in CI, with PASS/FAIL status and structured diagnostics; and
4. feed failures into a bounded LLM repair and re-validation loop.

Two public examples will be delivered: SAS -> R, using pregenerated SAS reference artifacts; and legacy R -> modern R, using the same framework.

Architecture

1. **LLM modernization interface** — model-independent generation and repair.
2. **Equivalence contract** — machine-readable definition of the behavior to preserve.
3. **Runtime/artifact layer** — executes the R candidate and normalizes candidate and reference outputs.
4. **Deterministic validator and evidence** — applies the contract, blocks acceptance on any required failure, and emits diagnostics and reproducibility evidence.

A thin R-friendly command or wrapper exposes the workflow without requiring users to touch orchestration internals.

Assumptions

- Trusted reference artifacts are available.
- Important behavior can be expressed through datasets and selected results, with acceptable differences (such as tolerances) stated explicitly.

- General-purpose LLMs can produce useful candidate transformations from source code and validation diagnostics.

If an LLM cannot complete a transformation, **equivR** can still validate a human-written or externally generated candidate; unsupported behavior is documented rather than treated as verified.

External dependencies

The project uses open-source R/Python libraries, standard CI tooling, and access to a general-purpose LLM, without depending on a particular vendor or model family. SAS is needed only to pregenerate the trusted reference artifacts; public CI will not require licensed SAS software. Proprietary model API costs will be self-funded or provided in kind.

Project plan

Start-up phase

Month 1. Finalize governance and MPL-2.0 licensing, define the first equivalence-contract format, implement the initial deterministic validator, and prepare the SAS -> R benchmark with pregenerated reference artifacts. By the end of Month 1, **equivR** should take a reference artifact, an R candidate result, and a contract, and return deterministic PASS/FAIL status with structured diagnostics.

Technical delivery

Milestone	Timing	Deliverable	Amount
M1	Month 1-2	Equivalence contract, deterministic validation, diagnostics, tests	\$1,250
M2	Month 1-2	SAS -> R modernization demonstration	\$1,000
M3	Month 2-3	Legacy R -> modern R, R-friendly invocation, CI, benchmark format	\$1,000
M4	Month 3-4	Model-independent bounded LLM repair loop and reproducibility evidence	\$750
M5	Month 4	Documentation, examples, release hardening, tagged release	\$1,000

Failure recovery. If an LLM cannot produce a correct candidate, we will narrow or manually repair the example while preserving the validation experiment. Unsupported SAS/R behavior will be documented as out of scope. Model-provider failure will not affect deterministic CI, and the repair loop stops after a bounded number of attempts.

Other aspects

Development will be public under MPL-2.0, with feedback and benchmark contributions through the repository. Phase 1 stays narrow: two small benchmark cases, artifact-based validation, and no live SAS in CI.

Budget & funding plan

Total requested: \$5,000. Milestone amounts are shown above.

No grant funds are requested for travel, workshops, hardware, publication fees, cloud services, AI credits, indirect costs, or licensed SAS software.

Success

Definition of done

Phase 1 is done when the SAS -> R and legacy R -> modern R workflows both run from a clean checkout, use explicit equivalence contracts, and produce deterministic diagnostics and reproducible validation evidence.

No live SAS runtime is required in CI. The repair loop is model/provider independent, and the LLM never determines acceptance.

Measuring success

- Both benchmark cases reproduce from a clean checkout, including SAS -> R using pregenerated trusted artifacts.
- Every required mismatch fails with actionable diagnostics.
- R-friendly local/CI invocation produces reproducibility metadata and machine-readable evidence.
- An initial public release is published.

The goal is not to show that an LLM can translate every analytical program. The goal is to show that modernization can be made verifiable: generation may vary, but acceptance remains deterministic.

Future work

The framework can extend to more SAS -> R and Python -> R cases, richer analytical result types, live runtime adapters, repository-scale modernization, and more automated repair, with the benchmark suite growing through community contributions.

R Consortium. 2026a. “Making R Submissions Reviewable for FDA.” 2026. <https://r-consortium.org/posts/making-r-submissions-reviewable-for-fda/>.

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